Dependent functions act on paths: for a section f of P there is a path from the transport of f(x) along p to f(y)

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled lem:mapdep (source lines 859-866); the proof span 868-880 is the author's first proof environment following it, retained line for line.

Editorial note. The author's second proof at the following lines is not counted and not retained on the card face.

Editorial note. The statement and proof cite no external labels, so no external references are declared.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the dependent action of a section on paths (apd). Only the counted lemma and its first author proof are retained; the author's second proof and the surrounding commentary remain uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent type families P over A, dependent functions (sections) f taking x to an element of P(x).
- Transport along a path: for p in x = y a function from P(x) to P(y), with its definitional behaviour on refl.
- The induction principle for identity types (path induction) with its conversion rule.

Lemma 0.0.1 (Dependent map)**.** *Suppose $f : \prod_{(x:A)} P(x)$; then we have a map*

$$\mathsf{apd}_f : \prod_{p:x=y} \big(p_*(f(x)) =_{P(y)} f(y)\big).$$

First proof. Let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family defined by

$$D(x, y, p) :\equiv p_*(f(x)) = f(y).$$

Then $D(x, x, \mathsf{refl}_x)$ is $(\mathsf{refl}_x)_*(f(x)) = f(x)$. But since $(\mathsf{refl}_x)_*(f(x)) \equiv f(x)$, we get that $D(x, x, \mathsf{refl}_x) \equiv (f(x) = f(x))$. Thus, we find the function

$$d :\equiv \lambda x.\, \mathsf{refl}_{f(x)} : \prod_{x:A} D(x, x, \mathsf{refl}_x)$$

and now path induction gives us $\mathsf{apd}_f(p) : p_*(f(x)) = f(y)$ for each $p : x = y$. □